120 Days of Quant Finance

Week 6: Stochastic Processes & Dynamic Asset Modeling
Day 41: Limits of the Black–Scholes Model and Motivation for Improved Models

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Introduction

The Black–Scholes model represents a foundational achievement in quantitative finance, providing a closed-form solution for option pricing under idealized assumptions. However, real financial markets exhibit behaviors that violate these assumptions, motivating the development of more advanced models.

Key Assumptions of Black–Scholes

The Black–Scholes framework assumes log-normal asset price dynamics with constant volatility, continuous trading, frictionless markets, and the absence of jumps. These assumptions simplify analysis but limit realism.

Volatility Smile and Skew

Empirical option prices imply volatility levels that vary across strike prices and maturities. This volatility smile contradicts the constant volatility assumption of Black–Scholes and reflects market perceptions of tail risk.

Fat Tails and Extreme Events

Observed asset return distributions exhibit fat tails and extreme events that occur more frequently than predicted by a Gaussian framework. As a result, Black–Scholes underestimates the probability of large losses.

Discontinuous Price Movements

Financial markets experience sudden price jumps due to information shocks and macro events. The continuous-path assumption of geometric Brownian motion fails to capture these dynamics.

Limitations of Dynamic Hedging

Black–Scholes relies on continuous delta hedging to eliminate risk. In practice, hedging occurs discretely and incurs transaction costs, leading to unavoidable hedging errors, especially during periods of market stress.

Motivation for Extended Models

The limitations of the Black–Scholes model motivate the development of models that incorporate stochastic volatility, jump processes, and multiple sources of risk. These models aim to better capture observed market behavior and tail risk.

Conclusion

Black–Scholes remains a critical benchmark in option pricing, but its limitations highlight the need for richer models. Understanding why the model fails is essential for selecting appropriate tools in real-world financial applications.